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ORIGINAL PAPER

Sentinel SAR-optical fusion for improving in-season wheat crop mapping at a large scale using machine learning and the Google Earth engine platform

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Abstract

In-season wheat growing area identification is of great importance for monitoring crop growth conditions and predicting related yield. In this study, we developed an approach to map wheat crops at a regional scale, using both the Synthetic Aperture Radar (SAR, Sentinel-1, S1) and Copernicus Optical (Sentinel-2, S2) satellite data, to estimate the extent of the wheat growing area. The approach relies on machine learning random forest classification algorithm performed in the Google Earth Engine (GEE) cloud platform. The methodology is based on three experiments, each consisting of the processing of a specific Sentinel time series imageries: a first experiment considering the S1 data solely, a second experiment with the S2 data solely and a third experiment with S1 + S2 data merged. The results showed that the third experiment combining SAR and optical data turned out with the best overall accuracy of 82.36% and a kappa coefficient of 0.77. These results indicate that the integration of Sentinel-1 and Sentinel-2 improved classification accuracy by 1.5 to 6% over the use of Sentinel-2 only. A comprehensive assessment based on survey samples revealed Producer and User accuracies of 84% and 81% respectively; and an F1-score of 0.82. The approach followed in the study provides a basis for mapping seasonal wheat areas that will support planning and policy decisions.

Keywords Winter wheat · Mapping · Sentinel · Crop classification · Random forest · GEE

Introduction

Agriculture provides most of the world's food and fabrics. It is the lifeblood of many African countries' economies, employing over 40% of the continent's total working population and accounting for almost one-tenth of the Gross Domestic Product (GDP) (Digital Earth Africa 2021). More than two-thirds of global wheat is used for food and one-fifth is used for livestock feed (Grote et al. 2020). Hence, in-depth and continuous mapping activities are essential to generate accurate spatial information and statistics on seasonal wheat-growing areas, to support crop monitoring and management activities, anomaly detection, damage assessment, and yield forecasting. Several methods are used to generate spatial and statistical information on wheat area for example manual sampling surveys estimates based on seed quantity and mapping approaches. Traditional methods based on field surveys are labor-intensive, time-consuming, and susceptible to subjective factors. To date, the democratization of satellite images, and the development of various methods of harnessing such data, have made it possible to map large-scale crop types and to estimate related areas, biomass, and yields using single or multi-sensor satellite data (Arvor et al. 2011; Kussul et al. 2015; Dong et al. 2020; Tufail et al. 2022).

Until recently, crop types mapping tasks using high-resolution images were mainly focused on the use of either optical or radar sensor images. Although promising, the combined use of high-resolution (10 m and beyond) optical

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and radar data is more accurate, compared to the use of a single sensor, especially concerning using machine-learning techniques for crop classification (Orynbaikyzy et al. 2019; Ibrahim et al. 2021; Mimouni et al. 2021; Valero et al. 2021). Jin et al. (2019) used the combined time series Sentinel-1 et 2 to map maize area and yield in Kenya and Tanzania. In 2018, Van Tricht et al. used joint Sentinel-1 radar and Sentinel-2 optical imagery to create a crop map for Belgium. They found that a combination of radar and optical imagery always outperformed a classification based on single sensor inputs. Tufail et al. (2022) used a machine learning random forest classification algorithm for accurate crop type mapping through the combination of SAR (S1) and optical (S2) time-series data, obtained results showed that combining all-weather accessible SAR and optical data skilled more precise outcomes. In addition, Valero et al. (2021) applied the S1-S2 classification system relating the so-called soft output predictions of two individually trained classifiers. The performances of the SenSAgri processing chain were evaluated in different agricultural systems. The results highlight the advantage of fusion for early crop mapping and the attention to sensing cropland areas before the identification of crop classes. These studies have investigated the ways of complementarity between optical and radar data, to allow the development of multisensory classification procedures that exploit both sources simultaneously over large spatial extents.

Previous researches on the use of remote sensing and machine learning in the context of wheat growing areas mapping have broadly highlighted several shortcomings, including among others the facts as follows: (i) wheat cultivated areas mapping could not be carried out before the end of the cropping season—during maturation stage, (ii) results of classification results could not be validated on different cropping seasons due to lack of ground sample data, (ii) complexity related to discrimination of wheat crop profile from other cereals, (iv) clouds and cloud shadows existence, leading to gaps in optical imagery and missing data in optical timeseries (CNCT and INGC 2012; OSS 2020), and (v) less consideration of vegetation phenology in crop growth mapping and monitoring, leading to issues in distinguishing and identifying some crop types such as cereals (Thiong'o et al. 2015; Wang et al. 2022), although knowledge about crops' phenology may play a decisive role in improving the results of cropland classification (Bargiel 2017; Kordi and Yousefi 2022; Nasrallah et al. 2018).

This study aims to generate a regional scale wheat map using machine learning classification algorithms. Our goals are twofold: firstly, to construct a comprehensive land cover map at the regional scale by effectively combining high-resolution optical and radar satellite imagery; and secondly, to accurately delineate large-scale wheat fields during the growth season. A key focus of this research is to develop a method that will be both accurate and simple enough to be easily adopted and further developed by agricultural technical institutions.

Material and methods

Study area

The study was conducted in the Governorate of Jendouba, which is in the extreme north-western part of Tunisia in the humid to sub-humid zone (Fig. 1). It covers an area of about 3102 Km^2. Jendouba Governorate is characterized by a uni-modal rainfall pattern lasting from May to October. Annual rainfall ranges from 400 to 1000 mm, while the mean monthly temperature for the past 35 years has ranged between 5 and 36°C. In the Jendouba, rainfed cereal growing occupies a prominent place, with wheat as the main crop type.

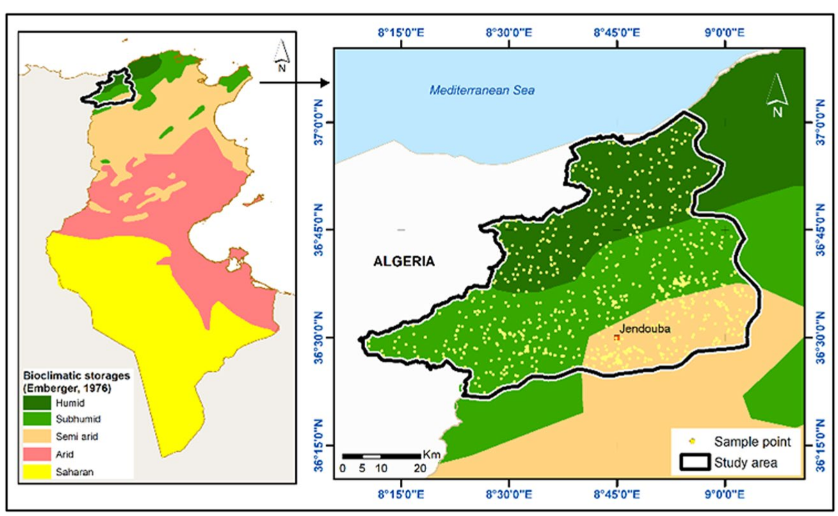

Fig. 1 Study area and distribution of sample points

Table 1 Distribution of validation and training field samples used in crop mapping (according to the ratio training 70% as training data and 30% validation)

Main types	Classes	Training	Validation	Total
Wheat	Wheat	504	216	720
Other seasonal crops	Barley, oats, maize, triticale, alfalfa, garlic, onions, fennel, tomatoes, peppers, etc	504	216	720
Perennial crops	Olive trees, oranges, lemons, mandarins, pomegranates, peaches, plums, strawberries, almonds, etc	504	216	720
Natural vegetation	Forests and rangelands (grassland, scrubland, garrigue, etc.)	504	216	720
Waterbody	Waterbody			
Built-up	Urban area			
Total		2016	864	2880

Data

Field data

Field surveys of the study area were carried out during the growing season of wheat crops from the 7th to the 20th of March 2022; a period characterized by good growth conditions of wheat sown area. Geo-referenced (GPS and camera) in situ data and photographs from sample crops' plots were collected using a random sampling approach. The data was regrouping different crops including wheat, barley, triticale, sugar beet, potatoes, olives, peaches, apples, rape, beans, alfalfa, garlic, onions, fennel, tomatoes, and peppers, other specific classes such as natural vegetation, water body, and urban area.

The data collected was compiled into a database which was structured into training and validation samples to make them usable for the hierarchical classification process and crop mapping tasks. It is composed of 2880 training samples for the study area (Table 1).

During the field survey, a broad crop calendar was generated at the level of the region. Due to the variability in climate or weather and other factors, there can be a shift in the timing of sowing and harvest of wheat over the years. The calendar (SI: Fig. S1) was utilized as a starting point to characterize the timing of the phenological stages of wheat.

Sentinel-1 data

Six Sentinel-1 images covering the study area were acquired from December 2021 to May 2022 at the rate of one image per month. These images were C-band instrument and Level-1 Ground Range Detected (GRD) Interferometric Wide Swath (IW) types. The images were characterized by dual-polarization (VV and VH) data at 10m spatial resolution (Table 2). Sentinel-1A&B ascending orbit was preferred as far as possible in this study because of the negative impact of morning dew on SAR observations. Indeed, the morning's dew absorbs C-band microwave radiation (Tufail et al. 2021; Hashemi et al. 2022). The data were pre-processed using the SNAP software. The pre-processing procedure included thermal noise removal, radiometric and geometric calibration, and speckle filtering. Lee filter was used for this research as previous studies showed that it gives better performance to reduce speckle noise compared to other filters (Lee et al. 1994, 2009; Punitham 2011).

Table 2 Sentinel-1 IW- GRD images data acquisition in Jendouba Governorate

No	DOY	Dates	No	DOY	Dates
1	337	03 December 2021	4	80	21 March 2022
2	20	20 January 2022	5	116	26 April 2022
3	56	25 February 2022	6	171	20 May 2022

Table 3 Sentinel 2 Level-2A, Bottom Of Atmosphere (BOA) imageries

No. of full scene	Acquisition dates	DOY	No. of full scene	Acquisition dates	DOY
1	28 December 2022	362	4	04 March 2022	63
2	28 January 2022	28	5	08 April 2022	98
3	06 February 2022	37	6	23 May 2022	143

Sentinel-2 data

To cover the entire study area, six Level-2A, Bottom Of Atmosphere (BOA) reflectance cloud-free images were collected from the Sentinel-2A/2B (Table 3). The acquired images ranged from December 2021 to May 2022 to make sure that most of the crop cycle would be tracked.

From Sentinel-2 images, we have selected 5 bands Blue (B2), Green (B3), Red (B4), Near Infrared (B8), and Shortwave Infrared 1 (B11) at 10-m resolution and Short Wave

Infra-Red 2 (B12) at 20-m resolution. The SWIR bands have been resampled to fit the resolution of the other bands. In addition to these bands, we have computed two sets of vegetation indices that have been extensively used in the literature for crop mapping: the Normalized Difference Vegetation Index (NDVI) and the Modified Soil And Vegetation Index (MSAVI).

Methodology

Overview of the main steps

In this study, optical and SAR data were utilized in three experiments in the process of mapping wheat growing areas. The flowchart of the detailed methodology is shown in Fig. 2. The description of the methodology is given in the following sections. Broadly, this method aimed at investigating, whether or not, the SAR when combined with optical data, can yield an accurate classification result. To get the results, a point-based samples random forest classification approach was performed, and the accuracy of the results was assessed later.

Random forest classification

Random forest was chosen as a supervised classification model to map the wheat area for reasons described in the introduction. The random forest is a nonparametric decision tree learning algorithm that is part of the machine learning classifiers (Breiman 2001). It is constructed using a bootstrapping technique in which each tree is fitted based on a random subset of training samples with replacement, while at each split in this tree, a random subset of the input features is also selected (Belgiu and Drăguţ 2016). It randomly selects a subset of the training sample through replacement to build a single tree by using a bagging technique where for every tree, data is sampled from the original complete training set. In this study, we have used the number of trees as 300. The selection was based on the hit-and-trial method. The number of variables highly affects the performance of a random forest classifier, which is usually set to the square root of the number of input variables. To determine the importance of variable, The GINI index was used. GINI index notes the class homogeneity and heterogeneity and is directly proportional to it; as the GINI index increases, class homogeneity increases and vice versa. The mean decrease in the GINI coefficient serves as a measure of variable importance. The zero GINI index shows the completion of splitting, and if a child node of the GINI index is smaller than the parent, it shows the split is successful (Watts et al. 2011; Tiwari et al. 2020).

Wheat area mapping

The wheat mapping was done using the random forest script within the GEE environment, which allows large-scale predictions at the pixel level. The random forest classifier was performed to assess the importance of each Sentinel-type

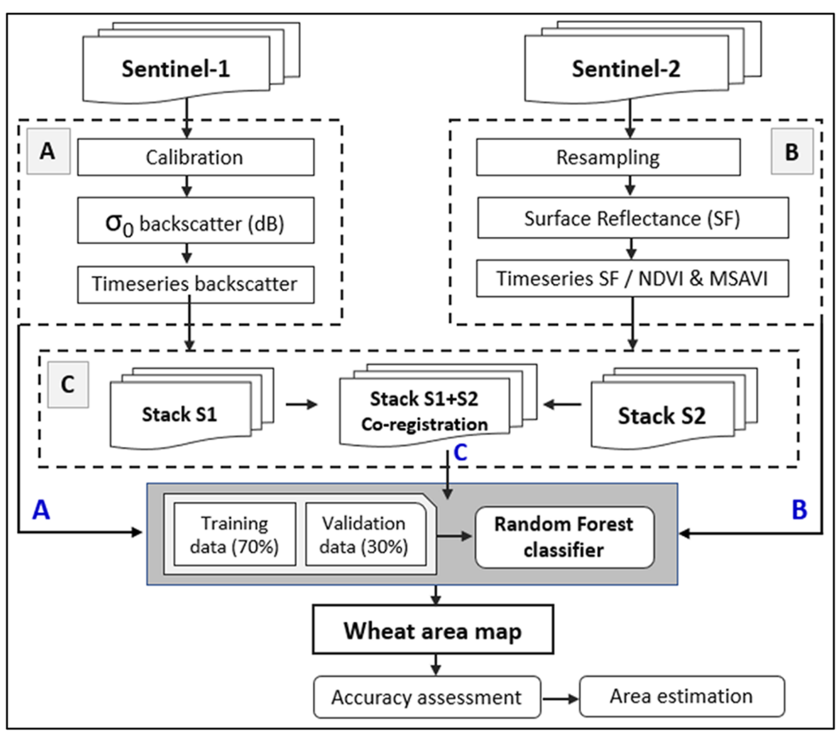

Fig. 2 Flow chart of methodological approach. Experiment-A: Sentinel-1 (IW-GRD; VV & VH) amplitude stack image, experiment-B: Sentinel-2 (Level-2A-BOA) reflectance stack image, and experiment-C: Sentinel-1 amplitude and Sentinel-2 (BOA) reflectance stack image

satellite data, including the vegetation indices, according to the three experiments:

- Sentinel-1 solely (experiment-A);
- Sentinel-2 solely (experiment-B);
- Sentinel-1 & and Sentinel-2 combined (experiment-C).

A two-step hierarchical classification procedure was followed, in which the first step distinguished between a general crop class and natural vegetation (forests, grassland, scrubland, garrigue, etc.), water, and built-up classes. In the second step, the crop classes (mainly seasonal and perennial) were further classified into different crop types. This hierarchical procedure is schematically outlined in Fig. 3. As indicated in the "Wheat crop mask generation" section, the optimal random forest parameters included the number of trees, the impurity criterion, the minimum number of samples required to split a node, the minimum number of samples required to be at a leaf node, and the maximum number of features to consider when looking for the best split. Classification of a sample resulted from the majority voting of class predictions across the trees in the forest.

Accuracy assessment

In this study, the accuracy assessment was conducted in two steps. Firstly, the results were checked by comparing with various ancillary data and trends on wheat cultivated areas from national agricultural statistical sources to identify gross errors. Secondly, the final data were used for quantitative accuracy assessment. As a reminder, from the field reference sample points, 70% were used for training and the remaining 30% for validation. Error matrices were generated for both Governorates separately and for each of the experiments A, B, and C. The performance of the different experiments A, B, and C was assessed using the main five different criteria and evaluation metrics, derived from the confusion matrices:

The overall accuracy (OA) evaluates the overall effectiveness of the classifier and is calculated as the total number of correctly-classifier pixels divided by the total number of validation pixels.

$$\mathrm{OA} = \frac{\mathrm{truepositive} + \mathrm{truenegative}}{\mathrm{truepositive} + \mathrm{falsepositive} + \mathrm{truenegative} + \mathrm{falsenegative}} \quad (1)$$

where

True positive is an outcome where the model correctly predicts the positive class;

True negative is an outcome where the model correctly predicts the negative class;

False positive is an outcome where the model incorrectly predicts the positive class;

False negative is an outcome where the model incorrectly predicts the negative class;

The precision or user's accuracy: The precision of class i is the fraction of correctly classified pixels concerning all pixels classified as this class i in the classification results.

$$\mathrm{Precision}_{\mathrm{i}} = \frac{\mathrm{Truepositive}}{\mathrm{Truepositive} + \mathrm{falsepositive}} \quad (2)$$

The recall or producer's accuracy: The recall for class i is the fraction of correctly classified pixels concerning all pixels of that ground truth class i.

$$\mathrm{Recall}_{\mathrm{i}} = \frac{\mathrm{Truepositive}}{\mathrm{Truepositive} + \mathrm{Falsenegative}} \quad (3)$$

The kappa coefficient is a statistical measure reflecting the difference between the actual agreement and the agreement expected by chance. Kappa is one of the most commonly used statistics to test interrater reliability (McHugh 2012; Kraemer 2015; Tang et al. 2015). If the kappa coefficient equals 0, there is no agreement between the classified image and the

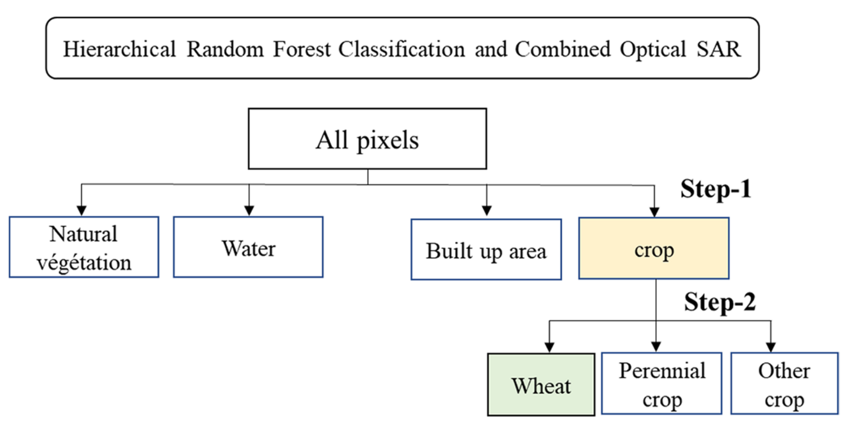

Fig. 3 Schematic overview of the two-step hierarchical classification procedure. In the first step, all crop types were combined into one general crop class, and in the second step, the focus was made on wheat

reference image. If the kappa coefficient equals 1, then the classified image and the ground truth image are identical. The higher the value of the kappa coefficient, the more accurate the classified map is (Tiwari et al. 2020).

$$Kappa = \frac{Accuracy of observed agreement - estimate of chance agreement}{1 - estimate chance agreement} \quad (4)$$

where

Accuracy of observed agreement $= \frac{\sum x_{ii}}{N}$, N = total number of observations, X_{ii} = observation in row i and column i,

Estimate of chance agreement $= \frac{\sum X_{i+} X_{+i}}{N^2}$, X_{i+} = marginal total of row i, and X_{+i} = marginal total of column i.

The F-score measures the accuracy of a class using the precision and recall measures, is defined as the harmonic mean of the precision and recall, and reaches its maximum value (best result) at 1 and minimum (worst result) score at 0.

$$\mathrm{F-score_i} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}} \quad (5)$$

Results

Land cover mapping in Jendouba Governorate

Pixel-based random forest classification technique was used to classify the images contained in experiment-A (Sentinel-1), experiment-B (Sentinel-2), and experiment-C (Sentinel-1 & Sentinel-2) for crop type mapping and wheat area identification.

Experiment A gave an overall accuracy of 89.88% and a kappa value of 0.75, whereas experiment B showed an increase in accuracy with an overall accuracy of 94.35% and a kappa value of 0.86 (Tables 4 and 5). The last experiment-C, which was the fusion of both types of optical and SAR data, demonstrated the most effective results with an overall accuracy of 95.88% and a kappa value of 0.90; showing an increment of 6% in the overall accuracy compared to experiment-A, and an increment 1.53% with experiment-B.

Table 4 Land cover mapping results based on RF classifier for the three tested experiments

Experiments	Data	Overall accuracy (%)	Kappa coefficient
Experiment-A	S1	89.88	0.75
Experiment-B	S2	94.35	0.86
Experiment-C	S1 + S2	95.88	0.90

Land cover maps

The first main result of this research is the land cover maps at the governorate level. Figure 4 highlights these land cover maps generated from the three experiments. The maps are composed of the 4 abovementioned thematic classes: "natural vegetation (forests, grassland, scrubland, garrigue, etc.)," "cropland," "built-up area," and "water body." The land cover map obtained from the experiment-C (fusion of optical and radar data) has recorded the highest results, with an overall accuracy of 95.88% and a kappa coefficient of 0.90 (Table 4).

Wheat area mapping using Sentinel-2 data

Usually in the study area, which is essentially characterized by rain-fed agriculture, there is a similarity in the cropping season and growth pattern between wheat and barley which tend to have the same growth patterns. Randomly collected training samples and vegetation indices were used to undertake the separation of wheat from barley crops during sowing, peak, and pre-harvest periods. Both NDVI and MSAVI thresholds were generated for the seasonal phenological cycles of both wheat and barley based on the crop calendar and collected sample points from the field. Figure 5 shows the growth patterns of wheat and barley in the Jendouba Governorates.

After examining the growth patterns of these sample crops, it was found that the wheat and barley crops are characterized by different values of NDVI and MSAVI during the growing period: a high NDVI value of barley during the 4 first months compared to wheat, knowing that both crops were conducted in rain-fed mode. However, it should also be noted that a high degree of overlap between the NDVI and

Table 5 Land cover map's accuracy from the confusion matrix

Classes	Experiment A			Experiment B			Experiment C		
	Producer accuracy	User accuracy	F1-score	Producer accuracy	User accuracy	F1-score	Producer accuracy	User accuracy	F1-score
Nat. veg	0.72	0.85	0.78	0.86	0.95	0.90	0.91	0.93	0.92
Cropland	0.96	0.90	0.93	0.98	0.94	0.96	0.97	0.96	0.96
Built-up	0.40	0.99	0.57	0.42	0.79	0.55	0.72	0.96	0.82
Waterbody	0.98	0.98	0.98	0.97	0.95	0.96	0.95	0.95	0.95

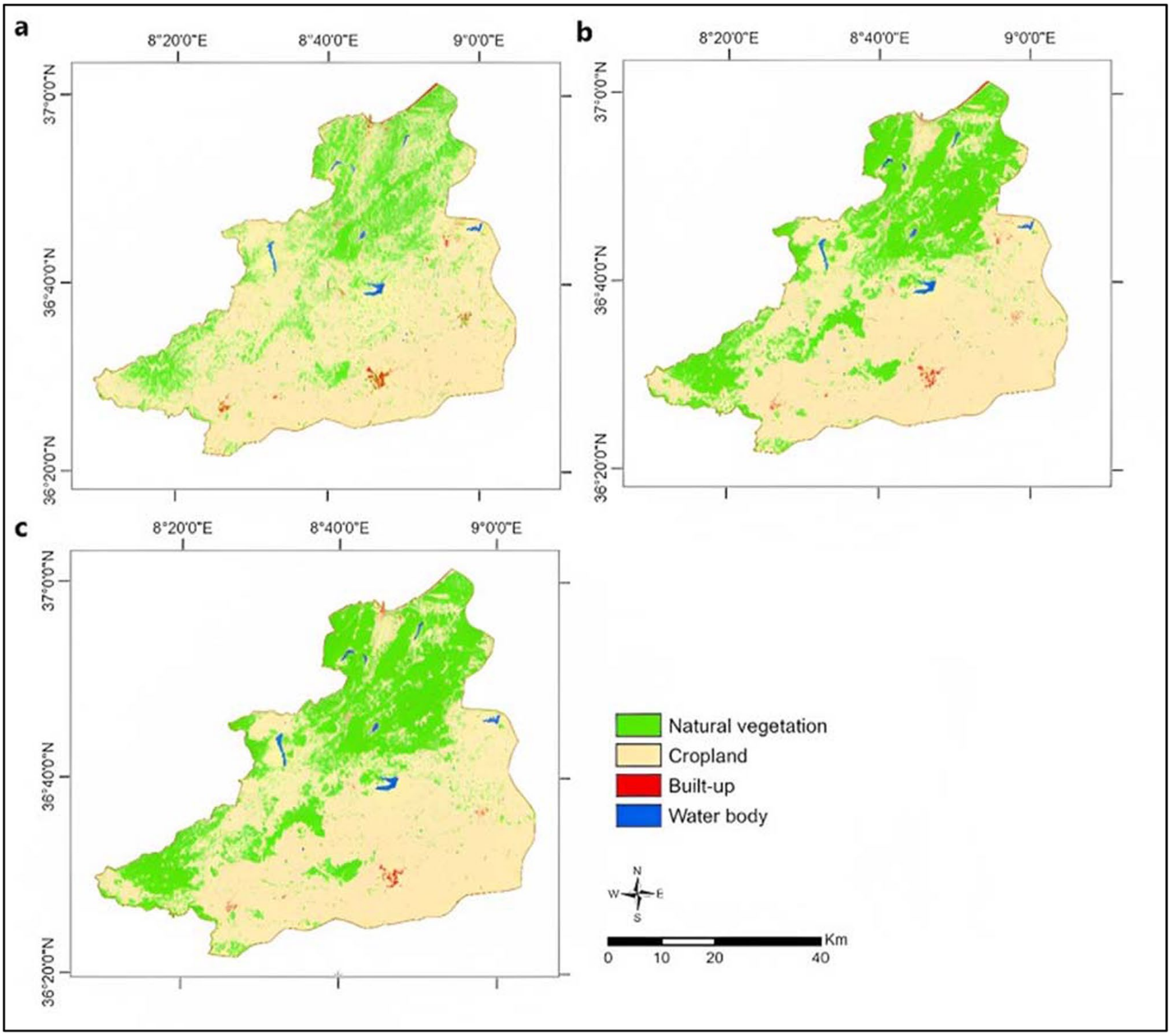

Fig. 4 Classified crop type maps for Jendouba Governorate in Tunisia (May 2022). Experiment-A (**a**), experiment-B (**b**), and experiment-C (**c**)

MSAVI values of wheat and barley is observed during the harvest period. This variability of responses and the overlaps during some periods make the threshold-based separation between wheat and barley quite challenging.

Wheat area mapping using Sentinel-1 data

The contribution of Sentinel-1 data in the discrimination of wheat and barley crops in the study area was done using randomly collected training samples and the seasonal time series Sentinel-1 data. Sentinel-1 images backscatter characteristics were used to separate the wheat from barley crops between the sowing and the pre-harvest period time. To examine the backscattering coefficient Sigma nought (σ_0) characteristics of these two overlapping crops, time-series composite Sentinel-1 SAR data from November 2022 to May 2023 were utilized to study the response of their backscatter characteristics at different periods (Fig. 5c). Given that SAR backscatter varies according to plant structure (Veloso et al. 2017; Nasrallah et al. 2018; Dineshkumar et al. 2019; Sun et al. 2019; Valero et al. 2021), distinct signals may be observed from different crops. The results clearly show that the VV polarization has higher backscattering coefficients (between -12dB and -6dB) than VH backscattering (between -22 dB and -14 dB). Moreover, one of the most noticeable facts was the high sensitivity of SAR temporal behaviour to both winter wheat growth cycles. Unlike the Sentinel-2-based NDVI and MSAVI temporal series, the Sentinel-1 VV and VH polarization temporal profiles witnessed a fluctuation

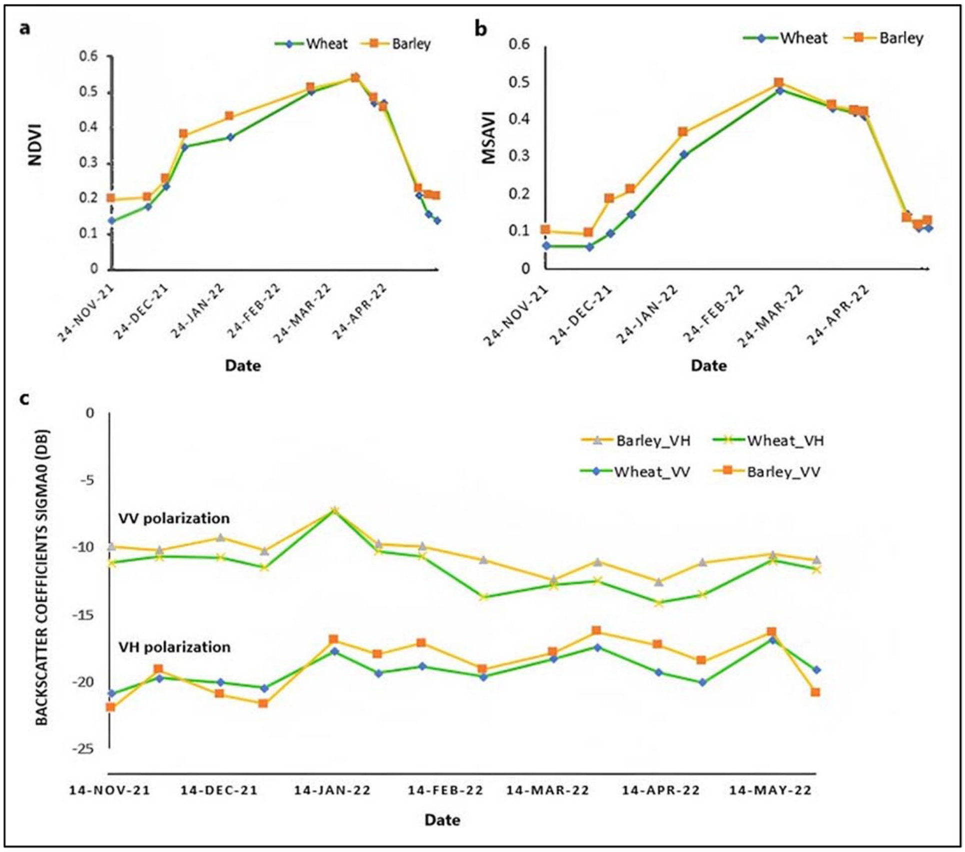

Fig. 5 Differences/similarities in the phenological cycles of wheat and barley, using NDVI (**a**), using MSAVI (**b**), and using backscatter coefficients VV and VH polarization temporal profile (**c**)

over this period. Within the 3 months (November to January) after sowing, the NDVI and MSAVI (Fig. 5) started increasing sharply just after the crop emergence, as it is therefore correlated to the greenness of the canopies (Labus et al. 2002; Duchemin et al. 2006; Wang et al. 2011). However, Sentinel-1 VV and VH polarization temporal profiles were experiencing fluctuation over this period. This is due to the high sensitivity of the Sentinel-1 backscatter to soil moisture, especially during this period which is marked by rainfall events needed from the germination to the heading stages of the crops (El Hajj et al. 2017; Baghdadi et al. 2018; Bazzi et al. 2019).

In summary, the results from the seasonal time series Sentinel-1 VV and VH backscatter characteristics in the Governorates of Jendouba showed high overlapping responses from the two crops (Fig. 5c). The optimal threshold-based separation is quite challenging using SAR backscatter datasets only remain a challenge. Random forest classification based on significant field sample data was used as an alternative to process the Sentienl-1 composite time-series backscatter data.

Wheat crop mask generation

The second main result of this research is the global crop map produced at 10 m of spatial resolution at the level of the governorate (Fig. 6a). This global crop map was derived from the land cover map produced from experiment-C which has given the better results among the three other experiments (overall accuracy of 95.88% and kappa coefficient of 0.90 (Table 4)). The global map broadly regroups three main classes: "wheat," "perennial crops," and "other crops" which regroups all the other seasonal crops. The wheat growing areas were highlighted in the map, showing that wheat is predominantly grown in the south-eastern part of the study area in winter, although it is generally scattered throughout the governorate.

From the global crop map (Fig. 6a), the seasonal wheat map was extracted and named wheat crop mask (Fig. 6b). This wheat crop mask is the final result of this research, generated from the outputs from the third experiment. The map shows the distribution of wheat (durum and soft) growing areas at the end of the winter season in May 2022. Moreover, it shows that wheat is the most dominant seasonal crop type in the study area. This dominance is in line with the interviews of the technical agricultural advisors from the CRDA the CTV and the local farmers during field visits held in March 2022.

Evaluation of the accuracy of the global crop map (regrouping the "wheat," "perennial crop," and "other crop" classes) was performed, based on the classification error matrix. The outputs are summarized in Table 6

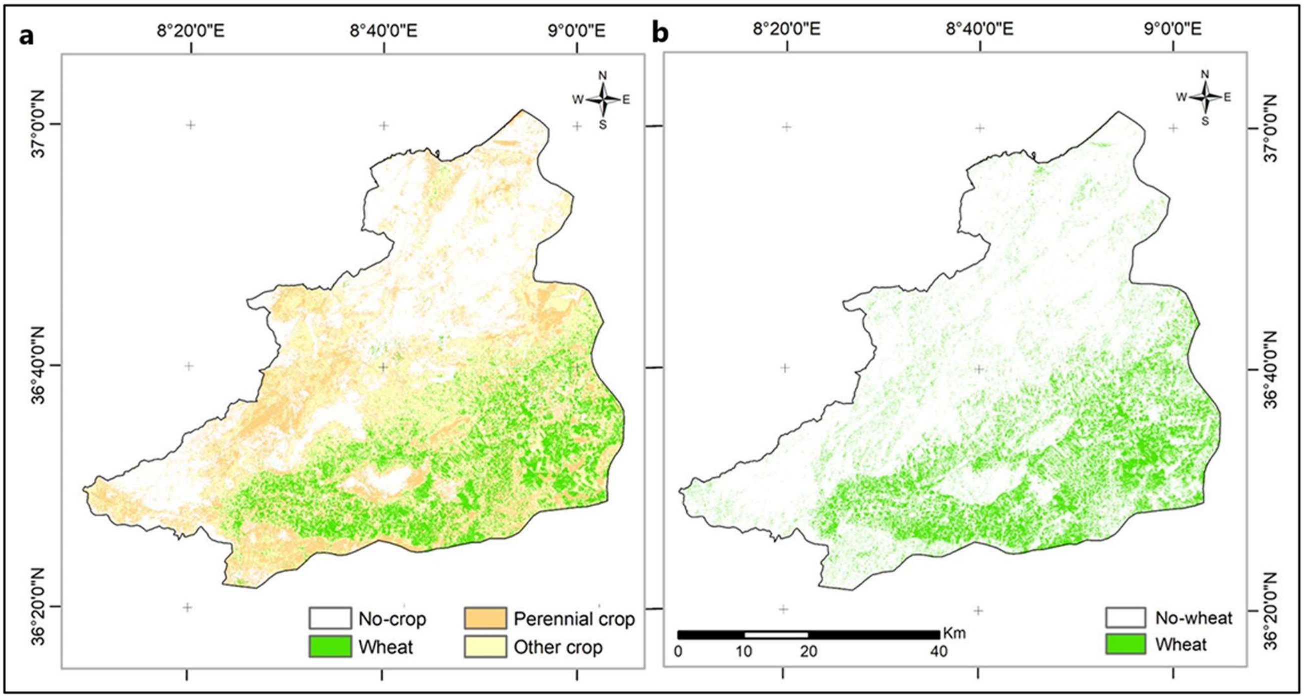

Fig. 6 Highlight on the wheat growing areas, generated respectively from the land cover and the global wheat maps (output from experiment-C) (**a**). Final wheat crop mask in the study area in May 2022 (**b**)

Table 6 Global crop mask accuracy—confusion matrix

Classes	Producer accuracy	User accuracy	F1-score
Wheat crop	0.84	0.81	0.82
Perennial crop	0.80	0.77	0.78
Other crop	0.72	0.74	0.73

which shows good results: overall accuracy, kappa coefficient, and class-wise accuracies all above 0.70. Generated at 10m of spatial resolution over an area greater than 300k ha, the wheat crop map which is the focus of this research is characterized by good class-wise accuracy (producer accuracy, user accuracy, and F1-score higher than 0.80).

The assessment procedure was essential before deriving confidently, the final wheat crop map confidently. To estimate the areas of wheat crops in the study area, we proceeded to convert the results of the wheat maps generated in the previous stages. We thus obtained 69,240 ha at the scale of the Governorate, which appears to show no discrepancy between the official statistics. The "Discussion" section gives more insights into this.

Discussion

Random forest for wheat mapping

The use of RF demonstrated a high performance of this method. This result is confirmed by several researchers who have explored the use of random forest for mapping wheat growing areas, achieving good accuracies through satellite image classification process. In this regard, Yao et al. (2022) compared the performance of random forest combined with deep neural network (RF + DNN) for crop classification, including wheat, using multi-temporal Landsat images. They found out that the RF + DNN combination outperformed the other methods (object-oriented and deep neural networks alone), achieving an overall accuracy of 98% and providing better discrimination of wheat fields. He et al. (2019) used the random forest algorithm in comparison with the attention-based long short-term memory (A-LSTM) algorithm to process time-series MODIS images and to map winter wheat with an accuracy of 82%. Abad et al. (2018) extracted vegetation indices and texture features from multi-temporal Landsat-8 data, training the random forest classifier to classify wheat growth stages with relevant accuracy (overall accuracy of 97% and kappa higher than 0.93). Liu et al. (2018) concluded that using the random forest algorithm and satellite images (Landsat-8 and China's Gao Fen 1 Wide Field of View (GF-1/WFV) at 16 m spatial resolution) can effectively improve wheat crop mapping (kappa coefficient of 0.85%). By mapping wheat growing areas using Landsat-8 and Sentinel-2 data, Basukala et al. (2017) and Chen et al. (2018) compared random forest, support vector machine, and maximum likelihood non-parametric machine learning algorithms and found that random forest achieved the highest overall accuracy (more than 80%) for wheat classification, indicating its suitability for mapping such type of culture. Neetu and Ray (2019) demonstrated the high performance of the random forest classifier (overall accuracy and kappa coefficient of 93.3% and 0.92 respectively) over two other machine learning classifiers: the classification and regression trees (overall accuracy and kappa coefficient values of 73.4% and 0.67 respectively) and the support vector machine (respectively 74.3% and 0.68). The studies have showcased the important potential of remote sensing-based data and indicators, machine learning, and classical techniques for accurate and timely mapping of wheat crops. The great majority of the authors have demonstrated that random forest achieved the highest accuracy and performed better than the other methods.

However, there are few studies (Thanh Noi et al. 2018; Amani et al. 2020; Dabija et al. 2021; Seydi et al. 2022) in which support vector machine (SVM) and convolutional neural network (CNN) have yielded best accuracies than other methods in wheat crop mapping. Most of the other research has highlighted the superior performance of random forest in wheat crop mapping tasks compared to other methods. In any case, it is essential to note that the choice of the most suitable method may depend on specific factors such as the context of the study area, data availability, and the specific objectives of the mapping task. The method also depends on the availability of training samples from field surveys and cloud-free time series satellite data for the entire growing season, which, according to Valero et al. (2019) and Dong et al. (2020), could make it somewhat challenging and restrictive for large-area crop mapping when such data are not available.

Optical and SAR data synergy

The study has revealed that optical and radar data can be effectively used to efficiently map land cover features and wheat growing areas on a large scale. The two experiments in this study, which involved the sole use of Sentinel-1 and Sentinel-2 data, have produced good results: overall accuracy being higher than 80% and a kappa coefficient of 0.75. Nevertheless, the step combining optical and SAR data turned out to have a much better overall accuracy of 82.36% and a kappa coefficient of 0.77. Several recent studies that have used this approach have reported higher accuracies compared to this work (Forkuor et al. 2014). They integrated high spatial resolution multi-temporal optical (RapidEye) and dual-polarized SAR (TerraSAR-X) data to map crops and crop groups in northwestern Benin using the random forest classification algorithm. They demonstrated that the use of SAR data improves accuracy by 15%.

Ofori-Ampofo et al. (2021) mapped crop types using the synergy between Sentinel-1 and Sentinel-2 with deep learning techniques. They confirmed that Combined Sentinel-1 and Sentinel-2 modalities are beneficial to increase classification performance in majority classes. They demonstrated that the fusion of high-level temporal features obtained from Sentinel-1 and Sentinel-2 time series produces reliable results when optical data is very limited.

Several other studies (Van Tricht et al. 2018; Tufail et al. 2021; Valero et al. 2021) confirmed the performance of the combination between SAR and optical data. As shown in a previous study (Inglada et al. 2016; Denize et al. 2019), our experiments also confirm that Sentienl-2 data discriminated croplands better than Sentinel-1. This may be because the SAR sensors are sensitive to vegetation structure, while optical sensors are sensitive to chlorophyll content. Thus, when observed at a 10m resolution, broad vegetation classes were discriminated more easily by their physiology than their physical structure. When using time-series optical and radar imageries in combination, a stack of spectral bands combined with NDVI and MSAVI data throughout the growing season has yielded sufficient optical descriptors for agricultural area mapping. The results also show that the method can be applied to successfully carry out seasonal and large-scale wheat crop maps in the Tunisian northern landscapes.

Wheat map

Accurately mapping wheat-growing areas at a large scale is a complex task. Several studies have utilized various approaches to classify satellite images and map wheat-growing areas in regions worldwide. The Asian region, especially China, has been a focal point for wheat mapping studies across various zones (districts, provinces, etc.). The accuracy assessment from these studies generally showed good results. In their research, Liu et al. (2017) used random forest classification to map winter wheat in the North China Plain using Landsat 8 imagery. The authors extracted various spectral and texture features from the imagery and achieved high accuracy in differentiating wheat from other land cover classes (overall accuracy of 92% and kappa coefficient value of 0.85). Dong et al. (2020) applied a phenology-based approach using a satellite-based vegetation index to produce an accurate 30 m winter wheat map (producer's and user's accuracies of 89.3% and 90.5% respectively) over the 11 key wheat producing provinces in China. In the Bekaa Plain in Lebanon, Nasrallah et al. (2018) applied the Simple and Effective

Wheat Mapping Approach (SEWMA) method, relying on 4 months of NDVI values. They reached an overall accuracy of 82.6% in 2016 and 87.0% in 2017.

These studies have highlighted various approaches that harnessed satellite images (optical and SAR) in several regions of the world to map wheat-growing areas with high accuracy. An important point to mention is that most of these methods are based on the use of vegetation indices derived from remote sensing data, particularly NDVI, which was used in almost all the other research. Authors such as Sun et al. (2019), Lopresti et al. (2015), have demonstrated in their research that NDVI is a valuable vegetation index, especially in wheat crop mapping studies. However, the mainstreaming of other indices such as the MSAVI, as well as raw satellite surface reflectance data in the classification process, may also contribute to improving the accuracy of the results. Moreover, compared to our study, several research studies have used very few amounts of in situ data in the training and validation processes when performing the wheat crop maps. The present study has taken these elements into account by integrating NDVI, MSAVI, and both Sentinel-1 time-series backscatter data and Sentinel-2 time-series surface reflectance images. The study has also widely used field sample data to train and validate the wheat map.

The accuracy assessment process in the abovementioned research was generally done based on the two main evaluation metrics: overall accuracy and kappa coefficient, which are certainly relevant. To gain more insight into our study, in addition to these two metrics, we have considered other criteria such as precision (or user's accuracy), recall (or producer's accuracy), and the F1-score, which are also useful for a thorough and exhaustive accuracy assessment (Olofsson et al. 2013, 2014).

Compared our result to the GEOGLAM-BACS global crop mask at a 5 km spatial resolution (Fig. 7), due to the mainstreaming of training samples from field surveys, the 10 m spatial resolution wheat map produced by this study appears to be more suitable and optimized for further activities such as crop condition monitoring and yield estimates.

The overall accuracy of the GEOGLAM-BACS global crop mask was based on a scoring criterion and ranged between 80 and 90%. Considering wheat crops, the current research resulted in an accuracy of 82.36%, which can be deemed relevant as it falls within the range considering appropriate for the GEOGLAM-BACS products. However, Oliva et al. (2018) reported a tendency to overestimate of cropland area values, which may be related to the coarser spatial resolution of this product and the issue of mixed pixels.

The result was then compared with the official records of the 2021–2022 agricultural campaign obtained from the *Direction Générale de la Production Agricole (DGPA)*. They

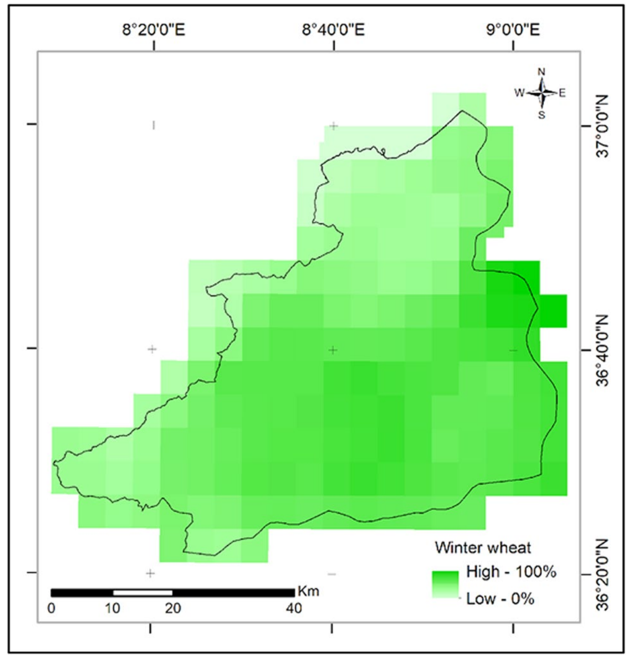

Fig. 7 Winter wheat crop at 5 km spatial resolution (GEOGLAM-BACS 2022)

Table 7 Distribution of cereal areas in the study zone

	Official stats (DGPA, 2022)	Experiment-C (2022)	Gap
	Area (ha)	Area (ha)	%
Jendouba Governorate	64 400	69 240	+7.5

reported a 7.5% gap, which seems low given the size of the Governorate (305,247 ha).

Table 7 illustrates the estimated surface area from the results of experiment-C, which, among the other three, showed the best results. There is no discrepancy between the statistics from experiment-C and the estimate from DGPA for the year 2022.

This research has emphasized the complementarity of optical and radar images to produce a 10m spatial resolution seasonal wheat crop map. Its particularities lie in the fact that (i) it has combined these high spatial resolution images, and (ii) was carried out over a large area leading to the generation of a large-scale wheat crop map. To our knowledge, there is currently no large-scale winter wheat map over Tunisia with high spatial resolution. Subject to the availability of field survey samples, the methodological approach that was adopted in this study offers possibilities of extrapolation and up-scaling of the map to other regions characterized by a similar bioclimatic context and storage. Subsequently, in future work, the main goal to be achieved is to improve the method, calibrate it, and upscale it to other governorates in Tunisia and even to the national scale. According to our findings, this approach of extrapolating maps on the wheat growing area to other regions remains largely unexplored in the research domain, and therefore, deserves further consideration.

Limitations

This research has demonstrated the advantage of leveraging optical Sentinel-2 data in complementary use with Sentinel-1 SAR data for agricultural lands and wheat growing area mapping. There are two limitations of the study. Firstly, the classification system is implemented in the Google Earth Engine platform, which requires some knowledge of coding to run the process online. Secondly, the reliable threshold-based separation of wheat from other crops has a limitation in the area where wheat is mixed with other crops that have similar phenology and plant structure. Though SAR images were used in synergy with optical data to help separate these crops from wheat, it requires substantial seasonal field samples which could be quite time-consuming and laborious, especially over large areas. An advantage of the approach implemented here is the application of dense time series satellite images over single-date images.

Conclusions

This study has unfolded a systematic methodology for wheat growing area mapping at the scale of the largest Tunisian administrative entity (the Governorate) with relevant potential for timely spatial information provision, which is essential for decision-making in support of food security. The study relied on a multi-step approach to leverage time-series optical and radar images to provide an estimate of the wheat cultivated area before the harvest period.

Previous research has demonstrated the complementary nature of optical and radar signals to improve classification accuracies. By applying random forest classification algorithms on time series data of both sensors during the agricultural season 2022–2023, this research has confirmed that the combination of radar and optical data could outperform a single-sensor classification procedure and lead to an increase in accuracy throughout the growing season.

Research findings have demonstrated the good performance of the complementarity and joint use of Sentinel-1 and Sentinel-2. They have thus highlighted the added value of a multisensory perspective. To date, there are no published distribution maps for winter wheat on a large scale such as Governorate with a high spatial resolution.

This research has contributed to filling this gap. It has produced a 10m spatial resolution winter wheat map of the Jendouba Governorate (regrouping 9 districts with mostly rainfed crops), which is expected to contribute to the timely monitoring of large-scale winter wheat growth and further, to timely yield estimate.

The adopted approach, while being replicable to ease the process of wheat growing areas monitoring on a near real-time basis, is also sustainable since it relies on Copernicus Sentinel data that are guaranteed up to the year 2030 and beyond. In future work, one of the main goals to be achieved could be the improvement of the method and its large-scale application to other staple or strategic crops (e.g., barley and olive).

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Declarations

Conflict of interest The authors declare no competing interests.

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